

ResidencyOS

Build Project Outline — Medical Physics Residency Management SaaS

How to use with Claude Code: Start each session: "We are building ResidencyOS. Here is the current state: [paste CLAUDE.md]. Today we are working on Phase X." Complete one phase at a time before moving to the next.

Phase 1: Foundation & Database Schema

- Set up GitHub repository with folder structure
- Create Supabase project
- Design and implement full PostgreSQL schema with all tables
- Set up Row Level Security (RLS) policies for multi-tenancy
- Seed database with test data (1 org, 1 program, 2 residents)
- Verify multi-tenant data isolation works correctly

Phase 3: Frontend Scaffold

- Initialize React app (Vite)
- Connect to Supabase
- Set up routing structure
- Build shared layout components (sidebar, nav, header)
- Build admin layout shell
- Build resident layout shell
- Deploy to Vercel/Netlify (get a live URL early)

Phase 5: Resident Management (Admin)

- Resident list view
- Add / edit resident
- Assign curriculum (auto-generates task instances)
- Application management (pending / approved / declined / inquiry)
- File uploads (transcripts, CV, references, letters)
- Inquiry log

Phase 7: Steering Committee (Admin)

- Committee member CRUD (name, role, active/inactive, dates)
- Meeting minutes CRUD (date, agenda, all fields)
- Attach members to a meeting
- File attachment for minutes
- Minutes list and detail view

Phase 2: Authentication & User Roles

- Configure Supabase Auth
- Implement three roles: Super Admin, Program Admin, Resident
- Build login / logout flow
- Build password reset flow
- Role-based routing (admin portal vs resident portal)
- Protect all routes by role

Phase 4: Program Management (Admin)

- Organization CRUD (create, read, update, delete)
- Residency Program CRUD
- Module CRUD with ordering
- Clinical Task CRUD per module
- Didactic Task CRUD per module
- Curriculum preview view (full program tree)

Phase 6: Resident Portal

- Resident dashboard with overall progress summary
- Curriculum view (modules to tasks)
- Clinical task progress (mark complete, add comments, upload docs)
- Didactic task progress (mark in progress / complete)
- Progress charts per module
- Handbook access (PDF viewer)

Phase 8: Module Evaluation

- Faculty creates evaluation per resident per module
- Evaluation form (competencies, reading, engagement, oral exam)
- Faculty signoff workflow
- Resident acknowledgment and comments
- Resident signoff workflow
- Evaluation status tracking (pending / approved)

Phase 9: Handbook (Dynamic PDF)

- Build handbook template (program info, curriculum, policies)
- Admin edits handbook sections in a rich text editor
- Generate downloadable PDF on demand
- Auto-update when curriculum or staff changes

Phase 10: Billing & Multi-Tenancy

- Stripe integration
- Subscription plans (monthly / annual)
- New organization self-serve signup flow
- Trial period logic
- Billing portal for customers
- Super admin dashboard (all orgs, status, usage)

Phase 11: Support & Notifications

- Transactional email (welcome, password reset, confirmations)
- In-app notifications (evaluation ready, task overdue)
- Support chat widget (Crisp or Intercom)
- Knowledge base / help docs structure

Phase 12: Polish & Launch Prep

- Mobile responsive review
- Loading states and error handling throughout
- Onboarding flow for new program directors
- Marketing landing page
- Custom domain setup
- Final security review
- Launch

ESTIMATED INFRASTRUCTURE COSTS (early stage)

Service	Purpose	Monthly Cost
Supabase	PostgreSQL + Auth + Storage	Free → \$25
Vercel / Netlify	Frontend hosting	Free → \$20
GitHub	Code repository	Free
Resend / SendGrid	Transactional email	Free → \$20
Stripe	Subscription billing	2.9% + 30¢/txn
Crisp / Intercom	Support chat	Free → \$25

REVENUE POTENTIAL (at \$199/month per program)

Customers	Monthly Revenue	Annual Revenue	Infra Cost
5 programs	\$995	\$11,940	~\$300/yr
10 programs	\$1,990	\$23,880	~\$600/yr
20 programs	\$3,980	\$47,760	~\$1,200/yr
50 programs	\$9,950	\$119,400	~\$2,400/yr

There are ~200 CAMPEP-accredited medical physics residency programs in the US. Capturing 10% = 20 customers = ~\$48K/year in recurring revenue.